xScape50 and xScape100 Product Range

User Manual

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<u>Controlled document note:</u> It is the user's responsibility to verify that the revision of any printed documentation matches the configuration controlled revision.

Document History

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2	2022-05-06	Updated section 5.1 – Specified sensor dependent parameters for integration time and line rate Updated section 5.2 – C_Smps1V0, V_Fpga1V0, V_Fpga1V8 and V_Fpga2V5. Updated section 12.5 Added section 12.6
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Table of Contents

List of Figures

List of Tables

List of Abbreviations

Abbreviation	Description
AIT	Assembly, Integration and Testing
API	Application Programming Interface
BAT	Block Allocation Table
BBT	Bad Block Table
CE	Control Electronics
CRC	Cyclic Redundancy Check
CSK	CubeSat Kit
DC	Direct Current
DN	Digital Number (usually refers to pixel value)
dTDI	Digital Time Delayed Integration
EGSE	Electrical Ground Support Equipment
EPA	ESD Protected Area
ESD	Electrostatic Discharge
FEE	Front-End Electronics
FPGA	Field Programmable Gate Array
GSD	Ground Sampling Distance
I2C	Inter-Integrated Circuit
ID	Identifier
IO	Input/Output
LVCMOS	Low Voltage Complementary Metal Oxide Semiconductor
LVDS	Low Voltage Differential Signalling
MSB	Most Significant Bit
N/A	Not Applicable
NC	Not Connected
OFE	Optical Front-End
PC	Personal Computer
PPS	Pulse Per Second
SNR	Signal to Noise Ratio
SPI	Serial Peripheral Interface
TDI	Time Delayed Integration
USB	Universal Serial Bus
VIS	Visible (spectrum)
VNIR	Visible and Near-Infrared (spectrum)

1. Introduction

1.1 Identification

Item Description: All products in the xScape50 and xScape100 Product Range, as per Table 1-1.

Table 1-1: Product Matrix

			Product Range	
			xScape50	**xScape100**
Product Type	**MonoScape**	**FM**	MonoScape50 052197	MonoScape100 055309
			MonoScape50 (with Enclosure) 055407	MonoScape100 (with Enclosure) 055410
			MonoScape50-CMV12000 IE 071470	MonoScape100-CMV12000 IE 071466
		EFM	MonoScape50-CMV12000 EFM IE 071451	MonoScape100-CMV12000 EFM IE 055418
			MonoScape50-CMV12000 EFM CE 071452	MonoScape100-CMV12000 EFM CE 055419
	TriScape	**FM**	TriScape50 046448	TriScape100 034293
			TriScape50 (with Enclosure) 055408	TriScape100 (with Enclosure) 055411
			TriScape50-CMV12000 IE 071471	TriScape100-CMV12000 IE 071467
		EFM	TriScape50-CMV12000 EFM IE 071453	TriScape100-CMV12000 EFM IE 055420
			TriScape50-CMV12000 EFM CE 071459	TriScape100-CMV12000 EFM CE 055421

Product Type	MultiScape CIS	FM	MultiScape50 CIS 046444	MultiScape100 CIS 036173
			MultiScape50 CIS (with Enclosure) 052912	MultiScape100 CIS (with Enclosure) 055412
			MultiScape50-CMV12000 IE 071472	MultiScape100-CMV12000 IE 071468
		EFM	MultiScape50-CMV12000 EFM IE 071460	MultiScape100-CMV12000 EFM IE 055425
			MultiScape50-CMV12000 EFM IE (with Filter) 071461	MultiScape100-CMV12000 EFM IE (with Filter) 055426
			MultiScape50-CMV12000 EFM CE 071462	MultiScape100-CMV12000 EFM CE 055427
	HyperScape	FM	HyperScape50 044550	HyperScape100 034647
			HyperScape50 (with Enclosure) 055409	HyperScape100 (with Enclosure) 055413
			HyperScape50-CMV12000 IE 071473	HyperScape100-CMV12000 IE 071469
		EFM	HyperScape50-CMV12000 EFM IE 071463	HyperScape100-CMV12000 EFM IE 055422
			HyperScape50-CMV12000 EFM IE (with Filter) 071464	HyperScape100-CMV12000 EFM IE (with Filter) 055423
			HyperScape50-CMV12000 EFM CE 071465	HyperScape100-CMV12000 EFM CE 055424

1.2 Intended Use

This document describes the safe operation of the xScape50 and xScape100 Product Range, hereafter also referred to as the "Imager", in its various modes. It also includes information on using the Simera Sense Electrical Ground Support Equipment (EGSE) with the Imager.

1.3 Context and Summary

The xScape50 and xScape100 Product Range includes electro-optical snapshot and push-broom imaging systems employing an AMS CMV12000 sensor. The xScape50 and xScape100 Product Range is produced by Simera Sense and is intended for earth observation applications. It is primarily designed to be implemented as part of an optical payload in a CubeSat. Its compact form factor allows for direct implementation into CubeSat structures; however, the xScape50 and xScape100 Product Range can also be used as part of larger satellite systems.

The operation and handling of the xScape50 and xScape100 Product Range, should be conducted by a qualified person who has experience in working with sensitive satellite components. Details regarding handling and integration of the xScape50 and xScape100 Product Range are presented in [9], [10] and [11].

Important information is distinguished in this manual using the following notations:

This is the safety alert symbol and is used to alert the user to damage that will be caused to the product if not adhered to. Obey all safety messages which follow this symbol to avoid damage to the product.

NOTICE A NOTICE indicates special precautions that must be taken to avoid degradation in performance of the product.

2. Applicable and Referenced Documents

Table 2-1 lists documents that are applicable, to the extent stated herein. In the event of conflict between the contents of the applicable documents and this document, the applicable documents shall take precedence.

Table 2-1: Applicable Documents

Ref. #	Reference
[1]	055227 – xScape100 Interface Control Document
[2]	055320 – xScape50 Interface Control Document
[3]	051715 – xScape Control and Data Interface Control Document
[4]	050443 – xScape SpaceWire Interface Control Document
[5]	051967 – xScape RS422,485 Interface Control Document
[6]	052024 – xScape USART Interface Control Document
[7]	055324 – xScape Image Data Compression Interface Control Document
[8]	053036 – xScape Data Encryption Interface Control Document
[9]	055404 – xScape100 Handling Instructions
[10]	059056 – xScape50 Handling Instructions
[11]	055311 – xScape-CMV12000 EFM Handling Instructions

Table 2-2 lists documents that are referenced in this document. In the event of conflict between the contents of the reference documents and this document, this document shall take precedence.

Table 2-2: Referenced Documents

Ref. #	Reference
[12]	DS000603 – AMS CMV12000 Datasheet

For undated references, the latest released version of the reference document applies. For dated references, subsequent versions of the document do not apply. It is best practice to always refer to the latest released version. Unless otherwise stated, web links referenced above were last accessed at the release date of the current version of this document.

3. System Overview

The xScape50 and xScape100 Product Range captures electromagnetic radiation, focuses the radiation on a sensor and converts the focused electromagnetic radiation into electrical signals. The xScape50 and xScape100 Product Range typically forms part of the payload of a satellite and is shown in the context of a typical CubeSat diagram in Figure 3-1 below.

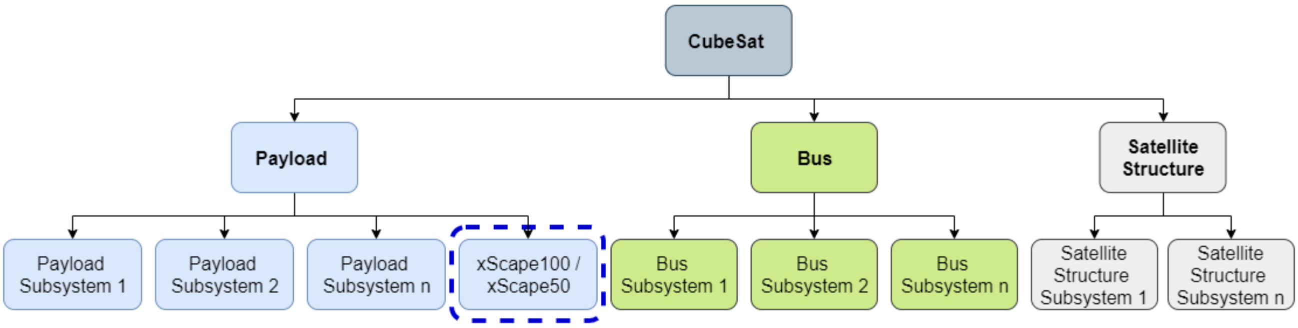

Figure 3-1: Typical CubeSat System Diagram

Each product in the xScape50 and xScape100 Product Range consists of three subassemblies which comprises of the xScape100 VNIR or VIS Optical Front-End (OFE), the Sensor Unit Assembly and the Control Electronics (CE). Figure 3-2 illustrates the physical imager system subassemblies and provides the axis definition of the xScape100 and xScape50 Imager products.

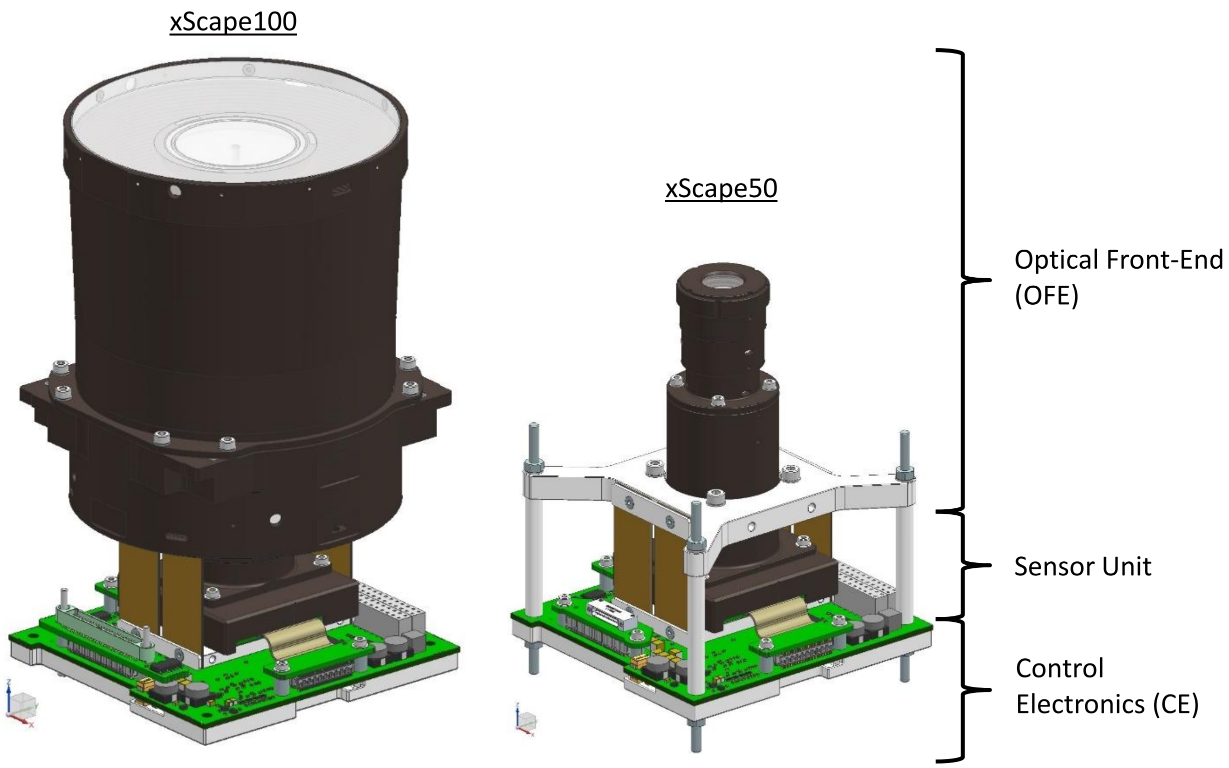

Figure 3-2: xScape100 & xScape50 Imager Subassemblies with Axis Definition

An exploded view of the Imager Sensor Unit and the Control Electronics is shown in Figure 3-3.

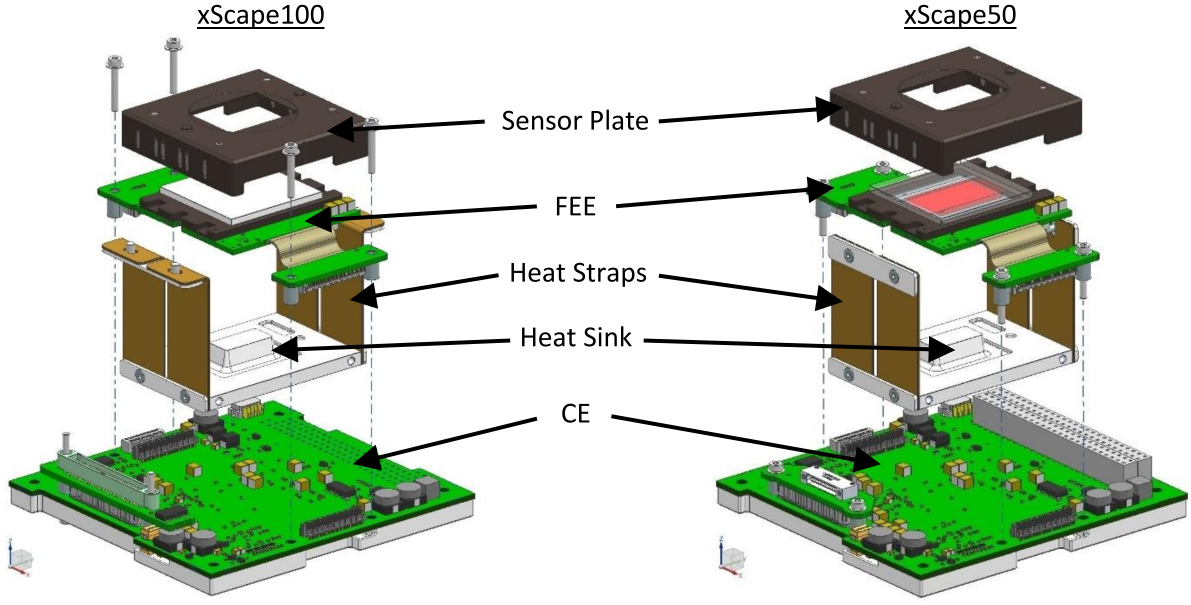

Figure 3-3: xScape100 & xScape50 Imager Sensor Unit and Control Electronics

In addition to the xScape100 and xScape50 Imager products, the xScape50 and xScape100 Product Range also includes the xScape EFM products: 'xScape-CMV12000 EFM Imager Electronics' and the 'xScape-CMV12000 EFM Control Electronics'. These are functionally representative of the Imager products in both electrical and software terms and can be used for functional and interface tests. Figure 3-4 shows an exploded view of the xScape-CMV12000 EFM Imager Electronics.

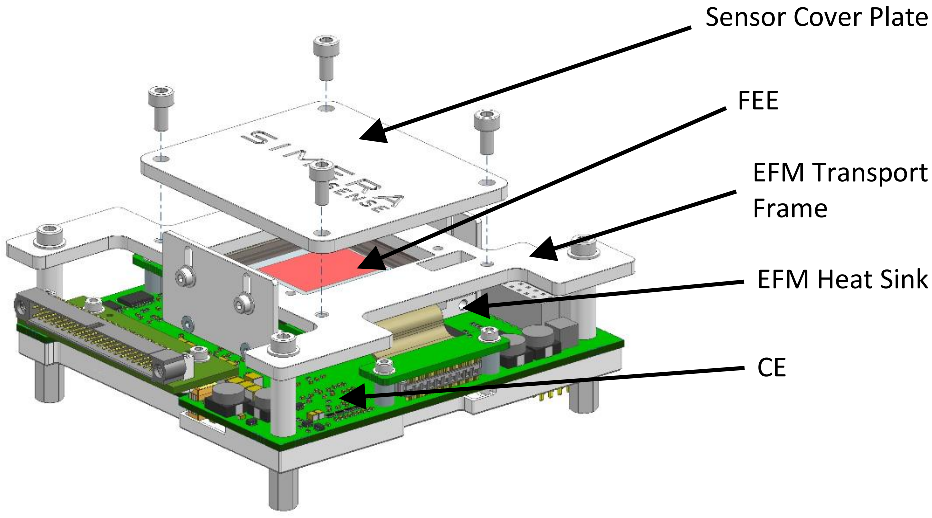

Figure 3-4: xScape-CMV12000 EFM Imager Electronics

Figure 3-4 shows an exploded view of the xScape-CMV12000 EFM Control Electronics, a reduced EFM consisting of only the Control Electronics and functional xScape firmware.

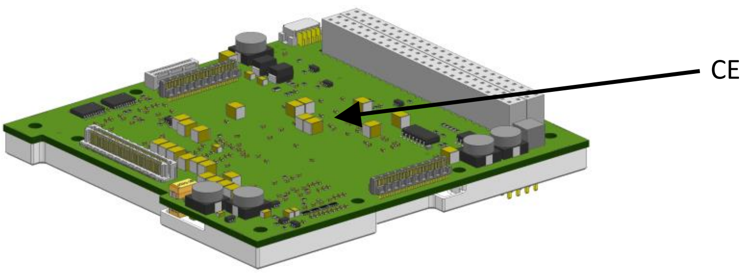

Figure 3-5: xScape-CMV12000 EFM Control Electronics

Each product in the xScape50 and xScape100 Product Range can be supplied with any of a number of high-speed breakout adapters. Harnesses are available for each breakout adapter to connect the Imager to the Simera Sense EGSE. Each product can also be supplied with a breakout harness for the CSK bus connector interface, if required.

Please refer to [1] and [2] for a detailed system and interface description.

4. Theory of Operation

The Imager is controlled and monitored through the Control Interface to capture images, perform optional image/data processing, and to read out the image data over its Data Interface. Refer to [3] for more information about the standard control and data interfaces. MonoScape and TriScape products capture snapshot images, either single frames or bursts of frames. MultiScape CIS and HyperScape products capture line scan images of programmable height (length) and in any combination of bands available for that product.

Once the Imager has started-up successfully, as described in Section 4.6, snapshot or line scan scenes may be captured and stored into a session. Existing sessions, containing previously captured scenes, may also be read out.

Section 4.9 describes the sequence of commands required to capture an image. A session refers to a period during which data may be stored to the non-volatile flash memory and as such only one session can be open/active at any given time. Pixel data from the sensor is first captured to a buffer. For snapshot imagers, the buffer allows a burst of images to be captured at a high frame rate before the image lines are stored in the flash memory with a pixel depth of up to 10 bits. For line scan imagers, the buffer is used to perform the digital TDI operations before the image lines are stored in the non-volatile flash memory with a pixel depth of up to 12 bits. To reduce storage space during line scan imaging and perform 1st-order band alignment, a hold-off period in-between the bands, allows the storage for each band to be enabled sequentially. The hold-off period is computed from sensor geometry alone and does not include any possible effects from satellite attitude jitter and/or smear. See Section 0 for more information.

The image data is stored in a packet-based format. During an imaging session, pixel data from each sensor line is formatted into individual packets. Additional packets are injected into the stream in real time, so that relevant ancillary data may be included with the image data. The user (satellite bus) can inject user ancillary data at his/her own discretion via the control interface. There is no fixed format/requirement for user ancillary data which may include attitude data, ephemeris data and timing information. The Imager automatically generates a set of ancillary packets to allow the exact Imager settings applied during the session to be stored with the image data. The imager sensor temperature is also injected periodically into these imager ancillary packets.

Section 4.10 describes the readout command sequence. During image data read out (via the Data Interface) the packetized image data is read out via the Data Interface as a continuous stream. That is, the individual image data packets do not need to be requested one by one; they are read out sequentially. The session's entire packetized data stream can be read out, or it can be filtered to pass only a sub-set of the session packets based on the type of data they hold. For instance, only thumbnail image data may be read out. Data can also be filtered to only read out a small part of the image. This is done based on image line numbers. The image data can also be encrypted during read out with 256-bit AES in Counter (CTR) mode using a key and initialization vector supplied via the Imager control interface before commencing read out. Encryption is only possible if the Imager includes the xScape Data Encryption option.

The FEE must not be powered for extended periods of time to ensure the sensor's operating temperature specification is not exceeded.

4.1 Snapshot Imaging

MonoScape and TriScape imager products can only perform snapshot imaging. MonoScape Imagers take single-band greyscale images, while TriScape Imagers are equipped with RGB Bayer filter arrays. They support single frame captures with a programmable exposure (integration) time or bursts of frames with a programmable exposure (integration) time, a programmable frame rate, and a programmable number of frames in the burst. The maximum frame rate for a burst of frames is dependent on the chosen exposure time.

For precise snapshot imaging, the Imager should point and track (aka 'stare' or 'dwell') the ground target as closely as possible. Take the Imager ground sampling distance (GSD) into account when calculating the required pointing stability and error margin. Doing so will minimize image smear caused by target movement during exposure (integration). When a burst of frames is captured, the Imager should track the ground target for the duration of the burst.

TriScape Imagers do not demosaic (debayer) imagery onboard the Imager; this step is performed after image data read out. The python code supplied with the EGSE can perform the demosaic step.

4.2 Line Scan Imaging

MultiScape CIS and HyperScape imager products can also perform line scan imaging (aka push-broom imaging) with a programmable line rate and a programmable number of lines to capture (i.e., the image height is programmable). The selected line rate value should equal the satellite ground speed divided by Imager GSD.

For precise line scan imaging, the Imager should point nadir while keeping the imager x-axis parallel with the direction of flight. The target must be in view and move in a straight line along the x-axis of the imager. Take both the Imager GSD as well as the maximum number of dTDI stages in any band into account when calculating the required pointing stability and error margin. This will minimize image smear caused by satellite pointing and stability error.

4.2.1 Enhancing SNR

The MultiScape CIS Imagers are line scan Imagers capable of performing TDI in the digital domain, aka dTDI. As the target is being scanned, the same line on the target is imaged N times, where N is the number of dTDI stages as programmed by the user. The 10-bit pixels output from the imaging sensor are accumulated to increase the Signal to Noise Ratio (SNR) by approximately factor $\sqrt{N}$. The number of dTDI stages, N, can be set independently for each spectral band using Imaging Parameter '0x32 Band Setup'. A fixed value, Imaging Parameter '0x35 Black Level', is subtracted from the $(N-1)$ repeat pixels to account for pixel bias to help avoid saturation due to accumulation for high values of N.

Because the imaging sensor outputs only 10-bit pixel values, the resultant 12-bit image saturation level depends on the number of dTDI stages selected as described by the following equation:

$$DN_{Saturated} = (2^{10} - 1) + [(N - 1)((2^{10} - 1) - Black\ Level)]$$

The saturation levels for various dTDI stages are provided in Table 4-1 . The standard value of 20 is assumed for the Black Level Imaging Parameter.

Table 4-1: DN(Saturated) vs Number of dTDI stages

N	$DN_{Saturated}$ (1)
1	1023
2	2026
3	3029
4	4032
>4	4096

(1) Black Level = 20.

Another method to increase SNR is to use "pitch control", also named Forward Motion Compensation (FMC). This can be implemented in addition to dTDI described above. FMC causes the perceived line rate to reduce by the pitch ratio, called the "FMC factor". Thus, if the FMC factor is 2, it means the perceived line period is increased by factor 2 which in turn means the integration time is also increased by factor 2 (approximately).

The graph in Figure 4-1 show the relative gains in SNR obtained by using dTDI and FMC individually. These SNR gains will compound when both dTDI and FMC are implemented together.

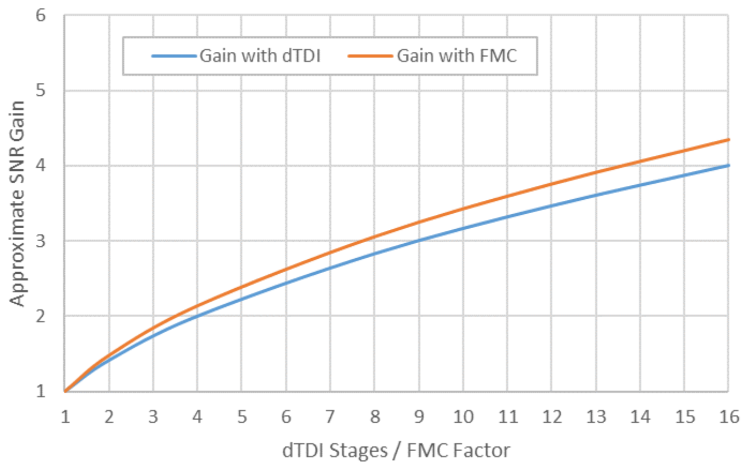

Figure 4-1: Approximate SNR Gain factor vs dTDI and FMC

4.2.2 Scan Direction

Imaging Parameter '0x34 Scan Direction' controls the direction in which the TDI operation should accumulate the image lines. The parameter is typically set once using the SET DEFAULT IMAGING PARAMETER command and not modified thereafter. The Imager axis definition is provided in the product Interface Control Document (ICD) along with a diagram showing the Imager ground projection that includes an arrow indicating the positive scan direction. It is important to set this parameter to the correct value depending on the Imager orientation relative to the satellite flight direction during line scan imaging.

4.2.3 Band Alignment

Since the filter bands are located some physical distance from each other, they do not image the same area on the target at the same point in time. Imaging Parameter '0x33 Band Start Row' contains the start row for each band. Use these values to calculate the number of rows between each band. Then use Imaging Parameter '0x31 Line Period' to calculate the difference in acquisition time between bands.

The platform pointing stability plays a big role in the band alignment, as the imager only performs a hold-off on the start of acquisition for each band based on only the sensor and filter geometry and does not take platform stability into account.

The band acquisition hold-off is performed by first calculating the number of sensor rows between the first band and each of the other enabled bands, called delta-rows. At acquisition start for the session, the imager acquires only the first band and pauses acquisition for the rest of the enabled bands until it has acquired the same number of rows (from the first band) that equals the band's delta-rows value calculated earlier. Thus, the bands will start to acquire the ground target one after the other; they start imaging the same area on the target at different times.

4.3 Ancillary Data

Data packets that do not contain image data are defined as ancillary data. The ancillary data is formatted into packets and stored to the imaging session in flash memory.

4.3.1 Imager Ancillary Data

The imager automatically generates ancillary data that:

a) Identifies the (platform) on which the Imager is integrated.
b) Identifies the imager (instrument) from which the data originated.
c) Provides imager configuration parameters used to acquire the image data.
d) Provides sensor temperature telemetry during the imaging session for calibration.
e) Optionally provides an absolute time reference. See 4.4.
f) Optionally provides accurate absolute time reference triggered by PPS pulses. See 4.4.

4.3.2 User Ancillary Data

The imager provides a framework to store other data to the imaging session which may contain attitude or other image related data. A centralised repository for all the imaging session data can thus be maintained and the User Ancillary Data is read out together with all other data in the session. The Imager accepts the User Ancillary Data through the STORE USER DATA command and automatically adds a CRC-32 checksum to protect the integrity of the data. The data is not timestamped or formatted in any way, although the first byte is reserved as an identification byte (ID) to distinguish between different, user defined, ancillary data packet types. Multiple User Ancillary Data packets of the same ID can be stored to a session. A maximum of 1024 bytes can be stored in each User Ancillary Data packet.

4.4 Time Management

The satellite bus is typically responsible for managing time on a satellite. This absolute time reference is referred to as the platform time. The Imager keeps a local 64-bit microsecond timer (unsigned integer) which starts from zero when the Imager is turned on. This timer value is referred to as the Imager time and is used as a time stamp when required for various packets. In order to synchronise the Imager time with the platform time, the satellite is responsible for sending the platform time to the Imager at the start of an imaging session. This platform time value will be stored as a TIME SYNCHRONISATION ancillary data packet, which also includes the Imager time value, so that the relationship between the two timers is known.

Some satellites include a Pulse Per Second (PPS) signal which may also be used by the Imager to generate additional TIME SYNCHRONISATION PPS packets containing the Imager timer value at the instant the PPS signal was asserted. These additional packets may be used to determine the relationship between the imager time and platform time more accurately. The Imager can also be configured to use the PPS signal as a trigger to initiate image capture. The PPS signal is configured using the SETUP PPS command.

4.5 States and Modes

A diagram of the Imager states and modes is shown in Figure 4-2.

NOTICE ‘xScape-CMV12000 EFM Control Electronics’ products are not fitted with Front-End Electronics and do not support Image Capture Mode.

NOTICE xScape EFM products, which includes ‘xScape-CMV12000 EFM Imager Electronics’ and ‘xScape-CMV12000 EFM Control Electronics’, implement a subset of the full health monitoring background tasks that excludes latch-up detection.

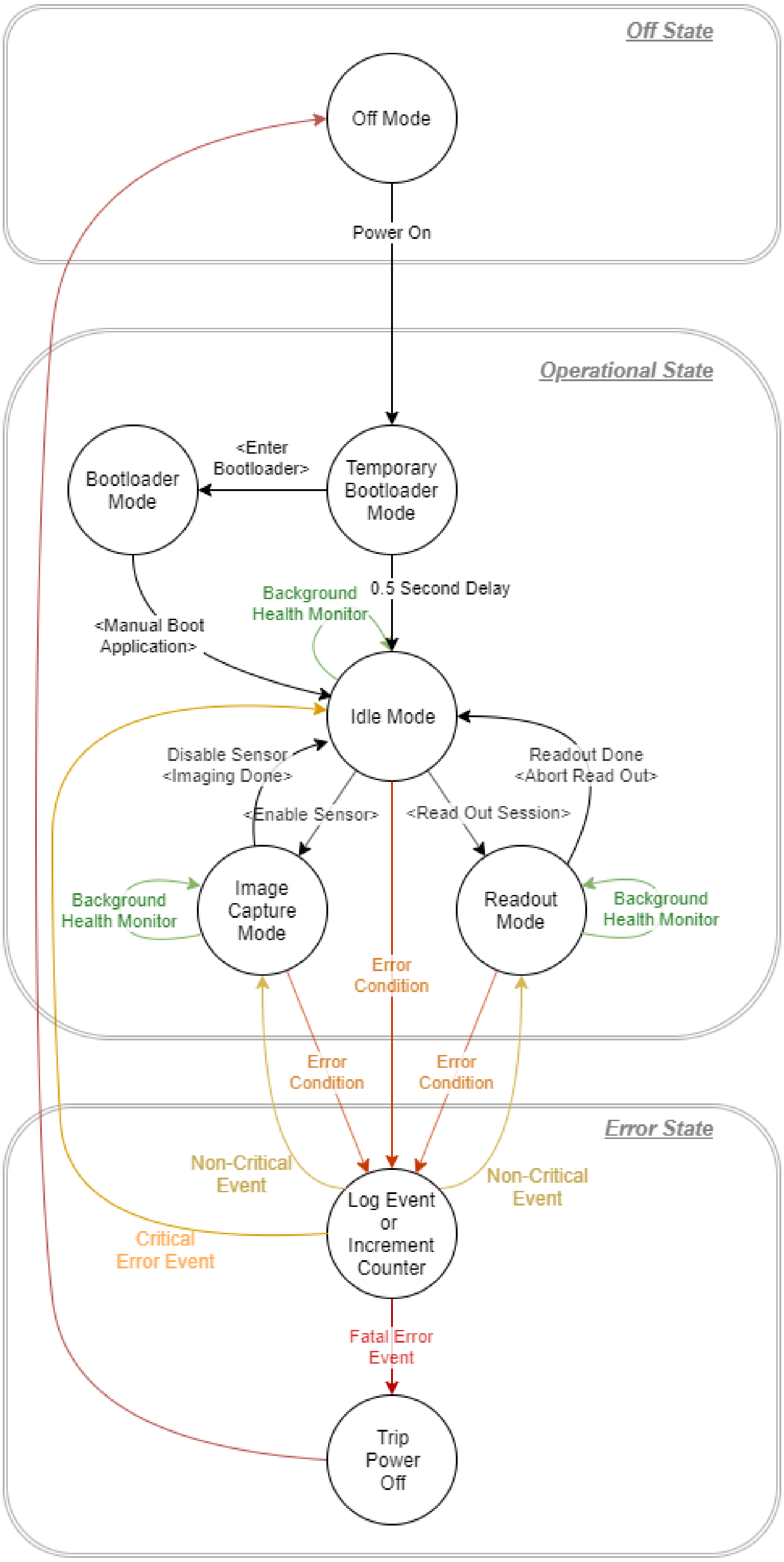

Figure 4-2: States and Modes Diagram

4.6 Start-Up Sequence

During start-up, after power is first applied to the Imager, the following actions are performed:

i) The Imager waits for 0.5 seconds in the bootloader. During this time the Imager can be requested to stay in the bootloader, else it will automatically load the Imager application and move to step ii.
ii) System parameters are loaded from non-volatile memory. Refer to Section 4.7 for more detail.
iii) Default Imaging Parameters are loaded from non-volatile memory. Refer to Section 4.8 for more detail.
iv) The NAND Flash session directory containing previously created sessions is scanned and loaded.

The start-up status can be queried using the START-UP STATUS request to determine whether the start-up was successful. The Imager start-up takes 1.0 to 1.3 seconds from the Off Mode in the Off State to Idle Mode in the Operational state.

4.7 System Parameters

The System Parameters are used to select, amongst others, the Imager Data Interface. The System Parameters are loaded from the Imager's non-volatile memory during Imager start-up. These (default) parameters can be changed by the user using the SET DEFAULT SYSTEM PARAMETER command, but it is not recommended to change these. The imager must be power-cycled or reset using the RESET command before the new system parameter value(s) will come into effect.

NOTICE Under normal operating conditions, it is not necessary to change the System Parameters, and it is advisable to leave them at their factory defaults.

The RESTORE SYSTEM PARAMETERS command can be used to do a factory reset of all the System Parameters. The imager must be power-cycled or reset using the RESET command before the factory system parameter values will come into effect.

4.8 Imaging Parameters

The Imaging Parameters control the way in which the Imager acquires image data. The Imager uses the parameters to configure the Imager for, amongst others, pixel binning, thumbnailing, pixel encoding, exposure time, snapshot frame rate, line scan line rate, and band selection.

Default values for all Imaging Parameters are loaded from the Imager's non-volatile memory during Imager start-up. These defaults can be changed by the user by issuing the SET DEFAULT IMAGING PARAMETER command. The updated default Imaging Parameter value(s) are only loaded at next start-up: after power-cycle or during reset using the RESET command.

4.9 Image Capture Sequence

A few command and request transfers (issued over the Control Interface) are required to capture and store an image scene. The following sections discusses the sequence for snapshots and line scan imagery.

NOTICE ‘xScape-CMV12000 EFM Control Electronics’ products do not support active imaging, but instead their non-volatile Flash memory is pre-loaded with test patterns and/or image captures that can be read out via the data interface.

NOTICE It is important that the user verifies that each command completes successfully (using the COMMAND STATUS request) before another command is issued.

4.9.1 Preparing for Image Capture

4.9.1.1 Optional: Set and/or Confirm Imaging Parameters

The default Imaging Parameters would have already been loaded during Imager start-up, but these can be changed as required to suit the particular imaging session’s requirements using the SET IMAGING PARAMETER command. The value of any Imaging Parameter can be confirmed (double-checked) using the GET IMAGING PARAMETER command followed by the IMAGING PARAMETER request.

4.9.1.2 Required: Open the Session and Configure the Imager

Issue the OPEN SESSION command to generate a Session Identifier and obtain the generated Session Identifier using the CURRENT SESSION ID request. The Session Identifier is required to identify the session (the stored image data) for future read out or deletion.

Issue the CONFIGURE command to configure the Imager. Verify that the Imager is configured, and that the session is open using the SUBSYSTEM STATES request.

4.9.1.3 Optional: Confirm Actual Hyperspectral Band Centre Wavelengths

NOTICE Only supported by ‘HyperScape’ products.

The nature of the sensor and hyperspectral filter array causes the actual band centre wavelength (CWL) to differ slightly (within 1 nm) from the value that was set.

The required CWL is programmed to Imaging Parameter ‘0x38 Hyperspectral Band Centre Wavelength’. While the imager is not configured, i.e., ‘Config State’ = ‘0’ as reported by the SUBSYSTEM STATES request, the Imaging Parameter values read back will equal the set CWL values. The Imaging Parameter can be read using the GET IMAGING PARAMETER command followed by the IMAGING PARAMETER request.

When the Configure command is issued, the imager calculates the closest CWLs possible as part of its configuration sequence. These CWL values are referred to as the ‘actual’ CWLs. Once configuration has completed successfully, ‘Config State’ = ‘1’ as reported by the SUBSYSTEM STATES request, and the Imaging Parameter values read back will equal the actual CWL values.

4.9.1.4 Required: Activate the Session

Issue the ACTIVATE SESSION command to reserve and erase the portion of flash memory needed. Use automatic mode to let the Imager calculate how much storage space is required (recommended) or set it manually. Then verify the session is active using the SUBSYSTEM STATES request.

At this point several image data packets are stored to the session in flash memory, such as SESSION START and IMAGER INFORMATION.

Issue the CURRENT SESSION SIZE request to determine how much flash memory storage space has been reserved, as well as the number of bytes already stored for the current active session.

4.9.1.5 Optional: Save an Absolute Time Reference

If it is desired to correlate the imager time with the platform time, issue the STORE TIME SYNC command to save an absolute time reference as a TIME SYNCHRONISATION image data packet into the session in flash memory. If a pulse per second (PPS) signal is present and enabled, each PPS pulse will inject a TIME SYNCHRONISATION PPS image data packet while the session is active. These two packets can be used to generate an absolute time reference with an accuracy of a few microseconds.

4.9.1.6 Required: Enable the Sensor

Up until now the FEE is still powered down to save power and keep the sensor cool. Issue the ENABLE SENSOR command to power up the FEE and prepare the sensor for image capture. The imager current consumption will increase at this point. Verify the sensor is enabled using the SUBSYSTEM STATES request.

4.9.2 Capture the Image

4.9.2.1 Required: Initiate Image Capture

Issue the CAPTURE IMAGE command to initiate the scene capture sequence: either immediately, or with a delay until the next PPS pulse. Image Data and Imager Ancillary Packets are now stored to the session in flash memory. Image data is saved on a line-by-line basis with each image line wrapped in a LINE DATA packet. Each snapshot in a burst is saved as a scene in the image data.

Monitor the image capture progress using the SUBSYSTEM STATES request. The imager will continue to inject TIME SYNCHRONISATION PPS image data packets at each PPS pulse, if enabled.

4.9.2.2 Optional: Store User Ancillary Data

Issue the STORE USER DATA command(s) to store user ancillary data into the Image Data stream while image capture is in progress (or thereafter). There is no fixed format for the user data. Using the 'User Packet ID' field may make it easier to identify different user data payload types or protocols.

4.9.3 Clean-up After Image Capture

4.9.3.1 Required: Disable the Sensor

As soon as the image capture is complete ('Capture State' returned to Idle), disable the sensor using the DISABLE SENSOR command to power off the FEE. By turning the FEE off as quickly as possible, the sensor will be provided more time to cool down for the next imaging session.

4.9.3.2 Required: Close the Session

Issue the CLOSE SESSION command to end the session and update the session parameters including the final number of bytes used. If the CLOSE SESSION command is not issued, e.g., if the imager is powered down during or right after image capture, no data will be lost, but an invalid session size might be reported. In these cases, the session status, as reported by the SESSION INFORMATION request, is marked as 'Not Closed'.

4.10 Image Data Read Out Sequence

The user may recall a previously captured session and read out the stored image data using the READ OUT SESSION command by following these steps.

NOTICE 'xScape-CMV12000 EFM Control Electronics' products do not support active imaging, but instead their non-volatile Flash memory is pre-loaded with test patterns and/or image captures that can be read out via the data interface.

NOTICE It is important that the user verifies that each command completes successfully (using the COMMAND STATUS request) before another command is issued.

4.10.1 Preparing for Read Out

4.10.1.1 Optional: Retrieve the Session Size in Bytes

Issue the GET SESSION INFORMATION command, followed by the SESSION INFORMATION request, to gather information about the session. This step is valuable if the receiver of the data needs to know beforehand how much data it needs to receive.

4.10.1.2 Required: Prepare the Receiver

Ensure the data receiver is ready to receive the data and that the necessary hardware flow controls are activated.

4.10.2 Read Out the Image Data

4.10.2.1 Required: Initiate Image Data Read Out

Issue the READ OUT RANGE SETUP command if partial image data readout is desired. Issue the READ OUT SESSION command, optionally specifying a filter of the data output. Data is now output onto the high-speed Data Interface.

Hardware flow controls, as implemented by the various data interfaces, are used by the receiver to throttle the data rate as required.

Monitor the image data read out progress using the SUBSYSTEM STATES request. Once the read out is complete (Read Out State returned to Idle), another Image Data Read Out sequence can be started.

4.10.2.2 Optional: Abort Read Out

The Read Out operation can be aborted at any point by issuing the ABORT READ OUT command.

4.10.3 Clean-up After Image Data Read Out

4.10.3.1 Optional: Delete the Session

Optionally delete the session once successfully read out by issuing the DELETE SESSION command. The imager also implements a function to delete all sessions from flash memory. Deleting all sessions will not reset the session identifier counter to zero but instead the counter will continue to increment for the life of the Imager.

4.10.3.2 Optional: Retrieve Session Status

Issue the SESSION INFORMATION request after session read out to retrieve the session status. If the "STO" bit is set, Flash ECC errors were detected during read out. See 4.12 for more information.

4.11 Confirming Imager Health

The Imager's health status can be attained by following the following steps in the sequence presented here.

4.11.1 Reset Reason

Issue the RESET STATUS request to retrieve the status of the Imager's most recent reset event. A 'Reset Reason' value of 1 or 64 indicate the Imager is healthy. If the 'Reset Reason' value is 128 it indicates a latch-up detection event and the 'Latch-up Flags' parameter will reveal the subcircuit(s) that caused the latch-up detection event.

4.11.2 Monitor Counters

The Imager monitors, and keeps count of, various error events. Most of these events are concerned with memory error detection and correction. Single bit errors are detected and corrected automatically and do not negate the Imager's health. The user can issue the MONITOR COUNTERS request to retrieve the count of errors detected since last start-up.

Watchdog timeouts and/or double bit errors mean the imager memory system is compromised as these are only detected and cannot be corrected. The Imager is automatically reset in these cases unless this has been disabled via the system parameters.

4.11.3 Telemetry with Sensor Disabled

With sensor disabled, request and confirm the CE telemetry channels are within range as specified in 5.1.6.1.

4.11.4 Startup Status

Issue the STARTUP STATUS request and confirm all status flags are zero.

4.11.5 Telemetry with Sensor Enabled

With sensor enabled, request and confirm both CE and FEE telemetry channels are within range as specified in 5.1.6.1 and 5.3.

4.12 Flash Memory Bad Block Management

The Imager uses NAND Flash memory for non-volatile storage of image data. The Imager maintains a bad block table to ensure non-functional Flash memory blocks are not used. The Imager bad block table is initialised at the factory and should only require updates after long-term use. The process is not automated but fortunately it is easy to implement.

4.12.1 Session Information

Issue the SESSION INFORMATION request after session read out to retrieve the session status. If the 'STO' flag is set, it means a Flash ECC error(s) was detected during read out.

4.12.2 Session Diagnostics

Issue the SESSION DIAGNOSTICS request to determine if there are any bad blocks detected during session operations. It also returns the start block and length of the session, in blocks.

4.12.3 Mark Blocks Bad

Take note of the session's range of blocks that may have errors. If this range of blocks present ECC errors again in the future, the blocks can be earmarked as bad. Mark the blocks as bad by issuing the SET BAD BLOCK command for each block in the range.

4.13 Diagnostics

The Imager provides three diagnostics requests to aid in low level debugging:

a) The SESSION DIAGNOSTICS request is useful to diagnose session errors in conjunction with the SESSION INFORMATION request. Also see 4.12.

b) The FLASH DIAGNOSTICS request is useful when diagnosing Flash Initialisation problems in conjunction with the START-UP STATUS request.

c) The SENSOR DIAGNOSTICS CHANNEL STATUS and SENSOR DIAGNOSTICS CHANNEL TIMING requests are useful when diagnosing problems with the sensor, when the ENABLE SENSOR command returns an error. Also see 5.5 and 5.6.

4.14 Bootloader

At power-up or after a system reset, the imager's processor first runs a bootloader, which then automatically boots the default application software after a period of 500 milliseconds (ms). During this initial 500 ms period, the user may suspend the automatic booting by issuing the ENTER BOOTLOADER command. The imager's protection features may also be disabled by issuing the DISABLE PROTECTION command during this initial 500 ms period.

The bootloader allows programming of two applications – application number 1 and 2. Application number 0 is the factory application, as delivered with the Imager, and cannot be re-programmed. Application 1 and 2 are available for future software updates from the manufacturer.

5. Product Specific Software Interface

The Imager provides a control interface as well as a data interface. Both these interfaces are required to fully operate the Imager. Command and Request transactions are placed onto the Imager control interface to issue commands and request telemetry. Image data is output via the data interface. [1] and [2] provide detail on the physical interfaces while [3] provides detail on the interface protocols generic to the xScape range of products. This section describes the interface protocols specific to the xScape50 and xScape100 Product Range.

5.1 Imaging Parameters

The following Imaging Parameter definitions and/or descriptions are specific to the xScape50 and xScape100 Product Range.

Parameter Number and Name		Parameter Value	
		Type	**Description**
Sensor Specific Parameters *<u>Note</u>: The Sensor Specific Parameters are typically not tuned/adjusted by the user*			
0x10	ADC Range	uint16	CMV12000 Register Address 116, bits 9:0
0x11	Bottom Offset	uint16	CMV12000 Register Address 87, bits 11:0
0x12	Top Offset	uint16	CMV12000 Register Address 88, bits 11:0
0x13	Gain	uint8	CMV12000 Register Address 115, bits 3:0
0x14	Black Ref. Enable	uint8	CMV12000 Register Address 89, bit 15
Snapshot Parameters			
0x20	Number of Frames	uint8	1: Single frame captured 2 to 128: Number of frames captured in a burst
0x23	Exposure Time	uint32	Integration Time (µs) Minimum value: 16 µs Maximum value: 16 seconds
Line Scan Parameters *<u>Note</u>: Only applicable to line scan products.*			
0x34	Scan Direction	uint8	Dependent on flight direction and Imager orientation. 0 : Imager scans in direction of $+x$ axis 1 : Imager scans in direction of $-x$ axis Note: This parameter is based on the fact that the Simera optics invert the image projected onto the sensor. If alternate optics are used which do not invert the image, the direction must be confirmed.

5.1.1 MonoScape100 and TriScape100

The following Imaging Parameter definitions and/or descriptions are specific to the MonoScape100 and TriScape100 Imagers.

Parameter Number and Name		Parameter Value	
		Type	Description
Snapshot Parameters			
0x21	Frame Interval	uint32	Inter-frame time (µs) for burst captures. The value indicates the time between frames in microseconds (inverse of the frame rate). Only applicable if Number of Frames > 1. Minimum value: Exposure Time + 6650 µs [1] Maximum value: 16 seconds

1. Maximum frame rate is 150 fps (with minimum exposure time of 16 µs).

5.1.2 MultiScape100 CIS

The following Imaging Parameter definitions and/or descriptions are specific to the MultiScape100 CIS Imager.

Parameter Number and Name		Parameter Value	
		Type	Description
Snapshot Parameters			
0x21	Frame Interval	uint32	Inter-frame time (µs) for burst captures. The value indicates the time between frames in microseconds (inverse of the frame rate). Only applicable if Number of Frames > 1. Minimum value: Exposure Time + 6650 µs Maximum value: 16 seconds
Line Scan Parameters			
0x31	Line Period	uint32	Line Period (us) (inverse of the line rate) Exposure Time (per TDI stage) is derived from this value as follows: Exposure Time = Line Period – 20 µs The line period is the same for all bands. Minimum value: described in section 5.1.2.1
0x32	Band Setup	uint8	Spectral band number. Counts from zero (0). Minimum value: 0 Maximum value: 7
		uint8	Enable band & number of dTDI Stages for this band. 0 : Disable the band 1 : Enable the band for line scan without dTDI 2 to 16 : Enable the band and perform dTDI, value equals number of stages.

5.1.2.1 Line Period Parameter

The Line Period parameter is the inverse of the line rate during line scan imaging. The minimum line period value is dependent on the number of spectral bands that are enabled, as follows:

Enabled Bands	Min. Line Period (us)
1	84
2	127
3	170
4	213
5	256
6	299
7	342
8	385

5.1.3 HyperScape100

The following Imaging Parameter definitions and/or descriptions are specific to the HyperScape100 Imager.

Parameter Number and Name		Parameter Value	
		Type	Description
Snapshot Parameters			
0x21	Frame Interval	uint32	Inter-frame time (µs) for burst captures. The value indicates the time between frames in microseconds (inverse of the frame rate). Only applicable if Number of Frames > 1. Minimum value: Exposure Time + 6650 µs Maximum value: 16 seconds
Line Scan Parameters			
0x31	Line Period	uint32	Line Period (us) (inverse of the line rate) Exposure Time (per TDI stage) is derived from this value as follows: Exposure Time = Line Period – 20 µs The line period is the same for all bands.

0x32	Band Setup	uint8	Spectral band number. Counts from zero (0). Minimum value: 0 Maximum value: 31
		uint8	Enable band & number of dTDI Stages for this band. 0 : Disable the band 1 : Enable the band for line scan without dTDI 2 to 8 : Enable the band and perform TDI, value equals number of stages. The number of dTDI stages selected affects the global dTDI processor limit, as explained in section 5.1.3.1.
0x3A	High-Accuracy Overhead	uint8	Overhead value for dTDI processing in high-accuracy mode. Typical value is 32. ** This parameter is Read-Only.*

5.1.3.1 Band Setup Parameter

The Band Setup parameter is used to enable and specify the number of dTDI stages for each spectral band. The number of dTDI stages selected per band is limited, as described by the parameter description above.

The Imager also places a global limit on the dTDI processing that is allowed for all bands, based on the dTDI stages selected per band as well as additional overhead per band.

The dTDI processing overhead is applied to hyperspectral bands (not the panchromatic band) when the imager is configured in high accuracy mode, based on the High Accuracy Mode parameter. In addition, the dTDI processing required per band is always rounded up to a multiple of 2.

The required dTDI processing is calculated as follows:

$$M_{dTDI}^{k=0} = 2 \times \left\lceil \frac{N_{dTDI}^{k}}{2} \right\rceil \qquad \text{(k = 0, for the panchromatic band)}$$

and

$$M_{dTDI}^{k} = 2 \times \left\lceil \frac{N_{dTDI}^{k} + N_{HA}}{2} \right\rceil \qquad \text{(k = 1..31, for hyperspectral bands)}$$

and

$$\sum_{k=0}^{32} M_{dTDI}^{k} \leq S$$

k is the spectral band number, ranging from 0 to 32,

N_{dTDI}^{k} is the number of dTDI stages selected (Band Setup parameter 0x32) for band k,

M_{dTDI}^{k} is the dTDI processing required (with overhead) for band k, which is always rounded up to a multiple of 2,

N_{HA} is 0 when configured in high band count mode,

N_{HA} is the high accuracy mode overhead (High Accuracy Overhead parameter 0x3A) when configured in high accuracy mode.

S is the dTDI processor limit.

The dTDI processor limit is calculated as follows:

$$S = \begin{cases} \lfloor (T_{line} \times 0.465) - 19 \rfloor, & T_{line} < 592 \\ 256, & T_{line} \geq 592 \end{cases}$$

where

T_{line} is the line period in microseconds (Line Period parameter 0x31)

As an example, with a line period of 667 µs (typical for a 500 km orbit height), the dTDI processor limit (S) is 256. In high band count mode, the user may set the panchromatic band as well as all 31 of the hyperspectral bands to 8 dTDI stages. The dTDI processing required may be calculated as follows:

$M_{dTDI}^{k} = 8$ for all 32 bands,

which means $\sum_{k=0}^{31} M_{dTDI}^{k} = 8 \times 32 = 256$ which is ≤ 256.

As a further example, in high accuracy mode, the user may set the panchromatic band to 5 dTDI stages, three hyperspectral bands to 5 dTDI stages and another three hyperspectral bands to 8 dTDI stages. With a typical High Accuracy overhead of 32, the dTDI processing required may be calculated as follows:

$M_{dTDI}^{k=0} = 6,\ M_{dTDI}^{k=1,2,3} = 38$ each, and $M_{dTDI}^{k=4,5,6} = 40$ each,

which means $\sum_{k=0}^{31} M_{dTDI}^{k} = 6 + (38 \times 3) + (40 \times 3) = 240$ which is ≤ 256.

5.1.4 MonoScape50 and TriScape50

The following Imaging Parameter definitions and/or descriptions are specific to the MonoScape50 and TriScape50 Imagers.

Parameter Number and Name		Parameter Value	
		Type	Description
Snapshot Parameters			
0x21	Frame Interval	uint32	Inter-frame time (µs) for burst captures. The value indicates the time between frames in microseconds (inverse of the frame rate). Only applicable if Number of Frames > 1. Minimum value: Exposure Time + 26600 µs [1] Maximum value: 16 seconds

1. Maximum frame rate is 37.5 fps (with minimum exposure time of 16 µs).

5.1.5 MultiScape50 CIS

The following Imaging Parameter definitions and/or descriptions are specific to the MultiScape50 CIS Imager.

Parameter Number and Name		Parameter Value	
		Type	Description
Snapshot Parameters			
0x21	Frame Interval	uint32	Inter-frame time (µs) for burst captures. The value indicates the time between frames in microseconds (inverse of the frame rate). Only applicable if Number of Frames > 1. Minimum value: Exposure Time + 26600 µs Maximum value: 16 seconds
Line Scan Parameters			
0x31	Line Period	uint32	Line Period (us) (inverse of the line rate) The line period is the same for all bands. Minimum value: 646 µs
0x32	Band Setup	uint8	Spectral band number. Counts from zero (0). Minimum Value: 0 Maximum Value: 7
		uint8	Enable band & number of dTDI Stages for this band. 0 : Disable the band 1 : Enable the band for line scan without dTDI 2 to 8 : Enable the band and perform TDI, value equals number of stages.

0x39	Exposure Time	uint32	Integration Time (µs) The maximum exposure time is limited by the Line Period. A value of zero will automatically calculate and set the maximum exposure time allowed. Maximum Value: Line Period – 630 µs Minimum Value: 16 µs

5.1.6 HyperScape50

The following Imaging Parameter definitions and/or descriptions are specific to the HyperScape50 Imager.

Parameter Number and Name		Parameter Value	
		Type	**Description**
Snapshot Parameters			
0x21	Frame Interval	uint32	Inter-frame time (µs) for burst captures. The value indicates the time between frames in microseconds (inverse of the frame rate). Only applicable if Number of Frames > 1. Minimum value: Exposure Time + 13300 µs Maximum value: 16 seconds
Line Scan Parameters			
0x31	Line Period	uint32	Line Period (us) (inverse of the line rate) The line period is the same for all bands. Minimum value: 1196 µs
0x32	Band Setup	uint8	Spectral band number. Counts from zero (0). Minimum Value: 0 Maximum Value: 31
		uint8	Enable band & number of dTDI Stages for this band. 0 : Disable the band 1 : Enable the band for line scan without dTDI 2 to 8 : Enable the band and perform TDI, value equals number of stages. The number of dTDI stages selected affects the global dTDI processor limit, as explained in section 5.1.6.1.
0x39	Exposure Time	uint32	Integration Time (µs) The maximum exposure time is limited by the Line Period. A value of zero will automatically calculate and set the maximum exposure time allowed. Maximum Value: Line Period – 1180 µs Minimum Value: 16 µs
0x3A	High-Accuracy Overhead	uint8	Overhead value for dTDI processing in high-accuracy mode. Typical value is 32. ** This parameter is Read-Only.*

5.1.6.1 Band Setup Parameter

The Band Setup parameter is used to enable and specify the number of dTDI stages for each spectral band. The number of dTDI stages selected per band is limited, as described by the parameter description above.

The Imager also places a global limit on the dTDI processing that is allowed for all bands, based on the dTDI stages selected per band as well as additional overhead per band.

The dTDI processing overhead is applied to hyperspectral bands (not the panchromatic band) when the imager is configured in high accuracy mode, based on the High Accuracy Mode parameter. In addition, the dTDI processing required per band is always rounded up to a multiple of 2.

The required dTDI processing is calculated as follows:

$$M_{dTDI}^{k=0} = 2 \times \left\lceil \frac{N_{dTDI}^{k}}{2} \right\rceil \qquad \text{(k = 0, for the panchromatic band)}$$

and

$$M_{dTDI}^{k} = 2 \times \left\lceil \frac{N_{dTDI}^{k} + N_{HA}}{2} \right\rceil \qquad \text{(k = 1..31, for hyperspectral bands)}$$

and

$$\sum_{k=0}^{32} M_{dTDI}^{k} \leq S$$

where:

k is the spectral band number, ranging from 0 to 31,

N_{dTDI}^{k} is the number of dTDI stages <u>selected</u> (Band Setup parameter 0x32) for band k,

M_{dTDI}^{k} is the dTDI processing required <u>(with overhead)</u> for band k, which is always rounded up to a multiple of 2,

N_{HA} is 0 when configured in high band count mode,

N_{HA} is the high accuracy mode overhead (High Accuracy Overhead parameter 0x3A) when configured in high accuracy mode.

S is the dTDI processor limit.

The dTDI processor limit has a fixed value: $S = 256$

For example, in high band count mode, the user may set the panchromatic band, as well as all 31 of the hyperspectral bands, to 8 dTDI stages each. The dTDI processing required may be calculated as follows:

$M_{dTDI}^{k} = 8$ for all 32 bands,

which means $\sum_{k=0}^{31} M_{dTDI}^{k} = 8 \times 32 = 256$ which is ≤ 256.

As a further example, in high accuracy mode, the user may set the panchromatic band to 5 dTDI stages, three hyperspectral bands to 5 dTDI stages and another three hyperspectral bands to 8 dTDI stages. With a typical High Accuracy overhead of 32, the dTDI processing required may be calculated as follows:

$M_{dTDI}^{k=0} = 6,\ M_{dTDI}^{k=1,2,3} = 38$ each, and $M_{dTDI}^{k=4,5,6} = 40$ each,

which means $\sum_{k=0}^{31} M_{dTDI}^{k} = 6 + (38 \times 3) + (40 \times 3) = 240$ which is ≤ 256

5.2 CE Telemetry

The nominal ranges for each of the CE Telemetry channels are shown below. Two ranges are specified, depending on whether the FEE/Sensor is disabled or enabled. If the ranges differ, they are highlighted. Certain telemetry channels are not used by the xScape50 and xScape100 Product Range and can be ignored.

NOTICE The telemetry limits are not defined for xScape EFM products and may differ.

NOTICE The telemetry limits are only applicable to the standard product offering, which excludes any additional options.

Channel Name	Mono-/TriScape				MultiScape CIS				HyperScape			
	Sensor Disabled		Sensor Enabled		Sensor Disabled		Sensor Enabled		Sensor Disabled		Sensor Enabled	
	Min.	Max.	Min.	Max.	Min.	Max.	Min.	Max.	Min.	Max.	Min.	Max.
V_FeeSmps	0	300	***2000***	***2060***	0	300	***2000***	***2060***	0	300	***2000***	***2060***
C_FeeSmps	0	25	***190***	***450***	0	25	***190***	***450***	0	25	***130***	***450***
C_FeeLdo	0	25	0	***55***	0	25	0	***55***	0	25	0	***500***
V_FeeNegSmps	-	-	-	-	-	-	-	-	-	-	-	-
C_Brd5V0	0	80	0	80	0	80	0	80	0	80	0	80
V_FeeOpAmp	-	-	-	-	-	-	-	-	-	-	-	-
V_SdramVtt	585	615	585	615	585	615	585	615	585	615	585	615
V_Brd3V3	3280	3400	3280	3400	3280	3400	3280	3400	3280	3400	3280	3400
V_Brd2V5	2490	2560	2490	2560	2490	2560	2490	2560	2490	2560	2490	2560
V_RefCal0	-	-	-	-	-	-	-	-	-	-	-	-
V_FeeLdo	0	10	***2950***	***3150***	0	10	***2950***	***3150***	0	10	***2950***	***3150***
V_IntTst0	-	-	-	-	-	-	-	-	-	-	-	-
V_REFM0	-	-	-	-	-	-	-	-	-	-	-	-
V_REFP0	-	-	-	-	-	-	-	-	-	-	-	-
C_BrdLdo	85	130	85	130	85	130	85	130	85	130	85	130
C_Smps3V3	90	220	***310***	***500***	90	220	***310***	***500***	90	220	***310***	***500***
V_Smps1V2	1190	1240	1190	1240	1190	1240	1190	1240	1190	1240	1190	1240
V_Smps1V0	990	1010	990	1010	990	1010	990	1010	990	1010	990	1010
C_Smps1V0	200	530	200	***650***	310	730	310	***850***	450	950	450	***1050***
C_Smps1V2	20	200	20	200	20	200	20	200	20	200	20	200
V_Brd1V8	1790	1840	1790	1840	1790	1840	1790	1840	1790	1840	1790	1840
C_SdramVtt	0	15	0	15	0	15	0	15	0	15	0	15
V_RefCal1	-	-	-	-	-	-	-	-	-	-	-	-
V_IntTst1	-	-	-	-	-	-	-	-	-	-	-	-
V_REFM1	-	-	-	-	-	-	-	-	-	-	-	-
V_REFP1	-	-	-	-	-	-	-	-	-	-	-	-
V_Fpga1V0	970	1030	970	1030	970	1030	970	1030	970	1030	970	1030
V_Fpga1V8	1745	1855	1745	1855	1745	1855	1745	1855	1745	1855	1745	1855
V_Fpga2V5	2425	2575	2425	2575	2425	2575	2425	2575	2425	2575	2425	2575
T_Fpga	-25	80	-25	80	-25	80	-25	80	-25	80	-25	80

5.3 FEE Telemetry

The FEE TELEMETRY request (refer to [3]) returns the telemetry as retrieved from the Front-End Electronics (which houses the sensor) by the GET FEE TELEMETRY command. The telemetry is only available while the Sensor is enabled by the ENABLE SENSOR command. Certain telemetry channels are not used, and their values can be ignored.

NOTICE The FEE TELEMETRY request is not supported by 'xScape-CMV12000 EFM Control Electronics' products.

FEE Telemetry			
Request ID	0x8B		
Data Bytes	29		
Channel Name	**Type**	**Description**	**Unit**
V_FeeSmps	uint16	1.3 V Supply Voltage	mV
V_FeeLdo	uint16	3.8 V Supply Voltage	mV
Reserved	uint16	-	-
V_TFLLow2	uint16	Sensor Bias	mV
V_TFLLow3	uint16	Sensor Bias	mV
V_Bandgap	uint16	Sensor Bias	mV
V_ResetL	uint16	Sensor Bias	mV
V_RefADC	uint16	Sensor Bias	mV
V_CmvRef	uint16	Sensor Bias	mV
Reserved	uint16	-	-
Reserved	uint16	-	-
V_IntTst	uint16	Not Used	-
V_REFM	uint16	Not Used	-
V_REFP	uint16	Not Used	-
T_CMV	int8	CMV12000 Temperature	°C

The nominal range for each telemetry channel is shown below.

Channel Name	Min.	Max.	Unit
V_FeeSmps	1980	2060	mV
V_FeeLdo	2950	3150	mV
Reserved	-	-	-
V_TFLLow2	40	55	mV
V_TFLLow3	40	55	mV
V_Bandgap	1125	1175	mV
V_ResetL	36	56	mV
V_RefADC	1750	1825	mV
V_CmvRef	530	585	mV
Reserved	-	-	-
Reserved	-	-	-

Channel Name	Min.	Max.	Unit
V_IntTst	-	-	-
V_REFM	-	-	-
V_REFP	-	-	-
T_CMV	-15	65	°C

5.4 OFE Telemetry

The xScape50 and xScape100 Product Range does not support the OFE TELEMETRY request.

5.5 Sensor Diagnostics Channel Status

The SENSOR DIAGNOSTICS CHANNEL STATUS request returns diagnostics information about the Imager's sensor. This information is only useful when diagnosing problems with the sensor, when the ENABLE SENSOR command returns an error.

Sensor Diagnostics Channel Status		
Request ID	0x94	
Data Bytes	9	
Return Data	**Type**	**Description**
Sensor Bottom Channel Status	uint32	Status for bottom sensor channels. Each bit represents a channel: Bits 0 to 31: Bottom Sensor Channels 0 to 31 If a specific channel failed, the bit will be set to '1'.
Sensor Top Channel Status	uint32	Status for top sensor channels. Each bit represents a channel: Bits 0 to 31: Top Sensor Channels 0 to 31 If a specific channel failed, the bit will be set to '1'.
Sensor Control Channel Status	uint8	Status for sensor control channel. If the channel failed, the value will be set to 1.

5.6 Sensor Diagnostics Channel Timing

The SENSOR DIAGNOSTICS CHANNEL TIMING request returns diagnostics information about the Imager's sensor. This information is only useful when diagnosing problems with the sensor, when the ENABLE SENSOR command returns an error.

Sensor Diagnostics Channel Timing		
Request ID	0x9D	
Data Bytes	65	
Return Data	**Type**	**Description**
Sensor Bottom Channel Timing	uint8 x 32	Timing value for bottom sensor channels. Each 8-bit value represents the timing value for one of the bottom 32 sensor channels.
Sensor Top Channel Timing	uint8 x 32	Timing value for top sensor channels. Each 8-bit value represents the timing value for one of the top 32 channels.
Sensor Control Channel Timing	uint8	Sensor control channel timing value. An 8-bit value represents the timing value.

5.7 Sensor Configuration Data Packet

The Sensor Configuration ancillary data packet is inserted automatically once the Imager is configured, but before an image is taken. Note that for the xScape50 and xScape100 Product Range the payload field is specific to the AMS CMV12000 sensor (Refer to [12]).

Sensor Configuration		
Packet ID	0xA2	
Flags	0x1	
Payload Length (bytes)	8	
Payload Field	**Type**	**Description**
ADC Range	uint16	CMV12000 Register Address 116, bits 9:0
Bottom Offset	uint16	CMV12000 Register Address 87, bits 11:0
Top Offset	uint16	CMV12000 Register Address 88, bits 11:0
Gain	uint8	CMV12000 Register Address 115, bits 3:0
Black Ref. Enable	uint8	CMV12000 Register Address 89, bit 15

12. Using the Imager with Simera Sense EGSE

The Imager can be connected to the EGSE via the Imager's control and data interfaces to perform functional tests. This section provides instructions on how to set up the Imager hardware with the EGSE and related software.

12.1 Required Items

Figure 12-1 shows a photo of EGSE items. Some items are not shown, as indicated in the text.

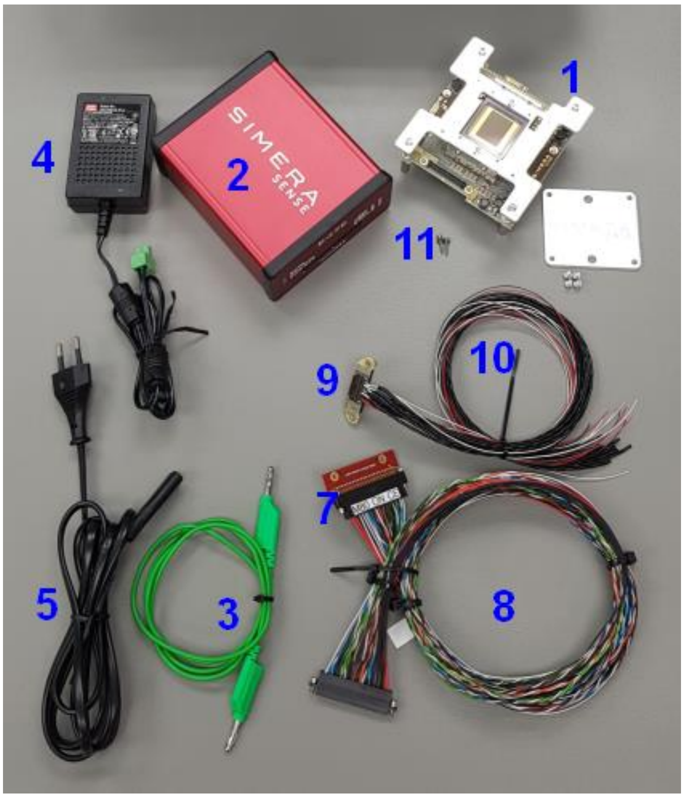

Figure 12-1: xScape EFM and EGSE Items

1. The assembled Imager. ('xScape-CMV12000 EFM Imager Electronics' is shown in Figure 12-1)
2. Simera Sense EGSE
3. EGSE Earth Cable, green, with banana connectors on both ends
4. Mains to DC Power Supply (Meanwell GST25B12-P1J)
5. Figure-8 to EU 2-pin power cord
6. USB 3.0 Cable (not shown in Figure 12-1)
7. Datamate Breakout Adapter (optional)
8. Datamate EGSE harness (optional)
9. Bi-Lobe Breakout Adapter (optional)
10. Bi-Lobe harness (optional)
11. 2 x M2 Socket Head Cap Screws with washers (optional)
12. Hex Key to fasten the M2 Socket Head Cap Screws (optional)

13. Personal Computer (PC) (not supplied)
    a. With Windows 10 installed
    b. With an available USB 3 port
14. RS422-to-USB cable (optional) (not shown in Figure 12-1)

12.2 Hardware Setup

⚠ Please adhere to the handling and operational precautions as stipulated in [9], [10] and [11].

⚠ Due to the static sensitive nature of the electronics, ESD protection measures shall be kept in place during use of the Imager.

1. Place all hardware items listed above on a static dissipative surface inside an ESD Protected Area (EPA).
2. Plug the EGSE earth cable into the EGSE earth banana socket. Connect the other end of the EGSE earth cable to the EPA earth.
3. Connect the DC Power Supply to the EGSE and plug the Power Supply's AC input into a wall socket. When a bench power supply is used, set the output voltage to 12 V and current limit to 1.0 A. Do not turn the supply on yet.
4. Connect the EGSE to a free USB 3.0 port on the PC, using the supplied USB cable. You should see the white EGSE 'STATUS' LED flash slowly.
5. If connecting via Breakout Adapter:
    a. The selected Breakout Adapter will already be installed onto the Control Electronics PCB.
    b. Connect the harness connector marked "EGSE" to the "Harness" port on the EGSE.
    c. Connect the other end of the harness to the breakout connector P9.
6. If connecting via CSK Bus Adapter:
    a. Connect the CSK Bus Breakout Harness Adapter to the Imager CSK bus interface connector H1 & H2.
    b. Connect the harness connector marked "EGSE" to the "Harness" port on the EGSE.
7. If the Imager is configured for RS422 control interface:
    a. Open Windows Device Manager
    b. Connect the RS422-to-USB to a free USB port on the PC.
    c. Note the COM number assigned to the RS422-to-USB converter
    d. Edit startup.py and enter the correct COM port number where indicated, near the top of the file.
8. Turn on the 12V DC Power Supply or turn on the bench power supply 12V output.

The hardware setup is now complete.

12.3 Software Setup

1. Install Python 3.7 on the PC
2. Clone the client git repository – the repository URL is shared with the client.
3. Open a terminal (command prompt) at the location where the scripts are placed, in the subfolder named `python`. This python folder will include the `startup.py` and `libusb-1.0.dll` files, as well as a subfolder named `simera`.
4. Ensure the following python libraries are installed. Administrative privileges may be required to install them.

Library Name	Command to install
`libusb1`	`pip install libusb1`
`numpy`	`pip install numpy`
`png`	`pip install pypng`

5. The following python library is required for TriScape Imagers. Administrative privileges may be required to install it.

Library Name	Command to install
`colour`	`pip install colour-demosaicing`

6. The following python library is required for Imagers that include the H.264 video option. Administrative privileges may be required to install it.

Library Name	Command to install
`ffmpeg`	`pip install ffmpeg-python`

7. Launch python (in the terminal) and run the `startup.py` script by executing the following command:

```
python -i startup.py
```

If no errors are produced, the software setup is now complete, and the user may begin interacting with the imager hardware via the EGSE.

If, however you receive an error about libusb1-1.0.dll not being found, please copy the dll to the location of your "python.exe" executable.

12.4 Using the Python Environment

Each time the user wishes to interact with the imager hardware, the following steps should be carried out:

1. Open a terminal (command prompt) in the `python` folder.
2. Launch python (in the terminal) and run the startup.py script by executing the following command:

```
python -i startup.py
```

The startup.py script will:

- instantiate the EGSE class, used to interface between the Imager and PC
- instantiate the Imager class, used to access the Imager API
- client specific Control and Data Interfaces are configured, if required
- provide 'ease-of-use' methods for image capturing, read out and image processing

3. Interact with the imager, using the methods in startup.py or the imager API via the xScape class

The following is an example of interaction with the imager. The table lists the key commands, and the terminal output is provided below.

Command	Description
`PwrOn()`	*'Ease-of-use' function.* Power on the Imager by enabling power switch in EGSE.
`imager.ReqStartupStatus()`	Check that the imager start-up was successfully - zero is returned for all the fields, indicating success
`imager.ReqStorageStatus()`	Query the available storage capacity - Imager reports 41/512 sessions and 4% capacity used

`SetupLinescanParameters( thumb=8, binning=2, lines=8000, period=667, bands=[16,8,0,0,0,0,0])`	*'Ease-of-use' function.*
	Configure the current (volatile) line scan parameters
	- Capture 8000 raw lines
	- 2x2 Binning applied. Resulting output image will have ½ the lines (4000) and ½ the raw line length.
	- 8x Thumbnail factor applied. Resulting thumbnail image will have 1/8 the lines (1000) lines and 1/8 the raw line length.
	- Line period set to 667 us (1500 Hz line rate)
	- Two Bands (of possible 7) are enabled
	  o Band 0 is setup for 16 dTDI stages
	  o Band 1 is setup for 8 dTDI stages

`CaptureAndReadoutLineScan( Info=True)`	*'Ease-of-use' function.* Capture, read out and generate PNG image files - Info is enabled which will print out the details of the image data received - The EGSE orange 'DATA' LED will blink while data is transferred via the Imager Data Interface
`PwrOff()`	*'Ease-of-use' function.* Power off the Imager

For this example, the python console output is as follows:

```
EGSE Loaded.
xScape Imager Loaded.
>>> PwrOn()
>>> imager.ReqStartupStatus()
(0, {'Busy': 0, 'Sys': 0, 'Img': 0, 'Sess': 0, 'Flash': 0})
>>> imager.ReqStorageStatus()
(41, 4)
>>> SetupLinescanParameters(thumb=8, binning=2, lines=8000, period=667, bands=[16,8,0,0,0,0,0])
>>> CaptureAndReadoutLineScan(Info=True)
Imager Configure Done.
Session ID 42 is now Active.
Injecting Time Sync Packet
Session has allocated 67108864 bytes (64 Mbytes) for storage in Flash
Image Capture Triggered...
Capturing 8000 lines at 667us line period. Duration is between 5.3 and 8.4 seconds.
PPS occured @ 1596020272.12
```

```
Image Capture Busy...
Injected User Ancillary Data @ 1596020272.49
PPS occured @ 1596020273.12
Image Capture Busy...
Injected User Ancillary Data @ 1596020273.49
PPS occured @ 1596020274.12
Image Capture Busy...
Injected User Ancillary Data @ 1596020274.52
PPS occured @ 1596020275.14
Image Capture Busy...
Injected User Ancillary Data @ 1596020275.49
PPS occured @ 1596020276.12
Image Capture Busy...
Injected User Ancillary Data @ 1596020276.50
PPS occured @ 1596020277.11
Image Capture Busy...
Session is now Closed.
Session Size after capture : 25923852 bytes of 67108864 allocated bytes used
Read Out from Session 42: 25923852 bytes
Reading out via Data Interface...
Read Out Complete and saved to line_session_42.bin
Exporting images to PNG files...
xScape Image Data
         1 Sessions:
                  SessionID         = 42
                  Closed            = True
                  ImagerInformation
                           ProductID         = 36173
                           SerialNumber      = 4
                           FirmwareVersion   = [2, 2]
                           SoftwareVersion   = [2, 7]
                  TimeSync
                           {'ImagerTime': 95162513, 'TimeFormat': 1, 'PlatformTime': 1596020271}
                           {'ImagerTime': 95693229, 'PPS': True}
                           {'ImagerTime': 96691710, 'PPS': True}
                           {'ImagerTime': 97690026, 'PPS': True}
                           {'ImagerTime': 98716472, 'PPS': True}
                           {'ImagerTime': 99695713, 'PPS': True}
                           {'ImagerTime': 100688125, 'PPS': True}
                  UserData
                           User Data ID 5:
                                             {'ExposureTimestamp': 5565050036, 'Data': b'This is sample
content for a User Ancillary Data Packet ID 5 at 1596020272'}
                                             {'ExposureTimestamp': 5581667372, 'Data': b'This is sample
content for a User Ancillary Data Packet ID 5 at 1596020273'}
                                             {'ExposureTimestamp': 5598348332, 'Data': b'This is sample
content for a User Ancillary Data Packet ID 5 at 1596020274'}
                                             {'ExposureTimestamp': 5613242556, 'Data': b'This is sample
content for a User Ancillary Data Packet ID 5 at 1596020275'}
                                             {'ExposureTimestamp': 5629859892, 'Data': b'This is sample
content for a User Ancillary Data Packet ID 5 at 1596020276'}
                  1 Scenes:
                           SceneNumber = 0
                           Type      = 1
                           Width     = 2048
                           Height    = 4000
                           2 RawBands:
                                    Band 0    = Single Band, 12-bit pixel depth
                                    Band 1    = Single Band, 12-bit pixel depth
                           2 ThumbnailBands:
                                    Band 0    = Single Band, 512 pixels per line @ 8-bit pixel depth
                                    Band 1    = Single Band, 512 pixels per line @ 8-bit pixel depth
Wrote 4000 image lines to '200729_125913.796031_42_0_Raw_0.png'
Wrote 4000 image lines to '200729_125913.796031_42_0_Raw_1.png'
Wrote 1000 image lines to '200729_125913.796031_42_0_TN_0.png'
Wrote 1000 image lines to '200729_125913.796031_42_0_TN_1.png'
Done.
>>> PwrOff()
>>>
```

12.5 Updating the Imager Application

The xScape Imager application software can be updated by following the steps below:

1. Ensure that the Imager is powered off.
2. Open a terminal (command prompt) in the `python` folder.
3. Launch python (in the terminal) and run the startup.py script by executing the following command:

   `python -i startup.py`

4. Power on the imager and enter the bootloader by calling the method:

   `PwrOn(Bootloader=True)`

5. Verify that the booloader is running by calling the following method:

   `PrintResetStatus()`

6. Ensure that the new application image file (*.img) is in the `python/bin` folder.
7. Program the new application image using the desired application number by calling the method:

   `ProgramApplicationImage(`*`appnum`*`,`*`"filename.img"`*`)`

   where *filename.img* is the name of the new application image file and *appnum* is the application number (1 or 2).
8. Wait until the programming is done. This may take up to 30 seconds.
9. Manually boot the new application by issuing the following command:

   `imager.ManualBootApp(`*`appnum`*`)`

   where *appnum* is the application number (1 or 2).
10. Verify that the application is running by calling the following method:

    `PrintResetStatus()`

11. Verify the new software version number by calling the method:

    `PrintImagerInformation()`

The python terminal should look as follows:

```
EGSE Loaded.
xScape Imager Loaded.
>>> PwrOn(Bootloader=True)
>>> PrintResetStatus()
Reset Reason   = 1 (Power-Up)
App Number     = 187 (Bootloader)
Run Time       = 26959 ms
>>> ProgramApplicationImage(1,"./bin/app_2_6.img")
Program Setup: App = 1
Programming Data... (please wait)
Programming Complete: Application 1, Version 2.6
>>> imager.ManualBootApp(1)
>>> PrintResetStatus()
Reset Reason   = 1 (Power-Up)
App Number     = 1 (User)
Run Time       = 6462 ms
>>> PrintImagerInformation()
    Product ID      : 36173
    Serial Number   : 4
    Firmware Version: 2.2
    Software Version: 2.6
```

12.6 Setting the Default Application

After programming a new application, it is possible to set the default application number, which will boot the specified application automatically at power-up, by following these steps:

1. Ensure that the Imager is powered off.
2. Open a terminal (command prompt) in the `python` folder.
3. Launch python (in the terminal) and run the startup.py script by executing the following command:

   ```
   python -i startup.py
   ```

4. Power on the imager and enter the bootloader by calling the method:

   ```
   PwrOn(Bootloader=True)
   ```

5. Verify that the booloader is running by calling the following method:

   ```
   PrintResetStatus()
   ```

6. Set the Default Application:

   ```
   imager.SetDefaultApp(appnum)
   ```

7. Power the Imager off:

   ```
   PwrOff()
   ```

8. Power the Imager on:

   ```
   PwrOn()
   ```

9. Verify that the application is running by calling the following method:

   ```
   PrintResetStatus()
   ```

10. Verify the new software version number by calling the method:

    ```
    PrintImagerInformation()
    ```

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